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Robust intrusion detection based on personalized federated learning for IoT environment

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ABSTRACT

In the dynamic and complex realm of the Internet of Things (IoT) and artificial intelligence (AI), it is a significant challenge to design a network intrusion detection system that balances accuracy, efficiency, and data privacy. Federated learning offers a solution by enabling the sharing of high-quality attack samples to enhance local models' intrusion detection capabilities without compromising local data privacy. However, most existing research on federated learning for intrusion detection assumes homogeneity among local models, which can reduce detection accuracy in real-world scenarios where local datasets are often non-independent and identically distributed (Non-IID). The Non-IID characteristic, marked by varied distributional properties and correlations, impacts model convergence and stability. To address this challenge, we propose a personalized federated cross learning framework (pFedCross) for intrusion detection, to manage imbalanced and heterogeneous data distributions. First, we present a collaborative model cross aggregation algorithm for personalized local model update, to solve the problem that one global model cannot always accommodate all the incompatible convergence directions of local models. Then, we introduce a gradient approximation α-fairness algorithm for global model generation to achieve a well-generalization. Finally, the experiments show that pFedCross outperforms baseline methods in improving model accuracy and reducing loss, highlighting its promise for enhancing IoT security.

1. Introduction

The Internet of things (IoT) (Nie et al., 2024) and artificial intelligence (AI) have become pervasive across diverse sectors, including smart homes, smart cities, and industrial automation. With the advent of Industry 4.0 (Iyer et al., 2024), data volumes are anticipated to surge, placing significant pressure on current centralized storage and processing systems. The proliferation of IoT devices with their inherent vulnerabilities, presents an expanded array of attack vectors and opportunities for cyber-criminals (Adam et al., 2024). Robust intrusion detection system (IDS) (Humayun et al., 2024) plays a vital role in identifying and countering malicious cyber-attacks in IoT networks. To address those security challenges, machine learning and deep learning techniques have been effectively integrated into IDS to achieve high detection accuracy. However, those methods often rely on centralized data processing, which requires data from all terminal devices to be transmitted to a central location for analysis. This not only increases data communication overhead and storage costs, but also aggravates privacy risks. Concerns about data privacy have led to the creation of data silos in industrial IoT, where data is withheld and not shared for analysis.

Data privacy protection is paramount in the evolution of IoT technologies. Conventional IDSs typically require the aggregation of data from IoT devices for model training. The apprehension over data leakage often leads organizations to restrict data sharing, thereby creating data silos (Rey et al., 2022). In response to these challenges, federated learning (FL) (Yurdem et al., 2024) has emerged as a promising alternative recently. FL allows for the collaborative training of models across decentralized clients while keeping the data localized, thus mitigating privacy risks and reducing the need for extensive data transfer and storage. This decentralized learning approach is particularly well-suited for the data privacy-preserving IoT ecosystems.

However, FL faces significant challenges due to the heterogeneity of data across different clients. In realistic scenarios, client data is often non-independent and identically distributed (Non-IID), contrasting with the IID assumptions prevalent in many research methodologies (Chiaro et al., 2023). Numerous studies have indicated that

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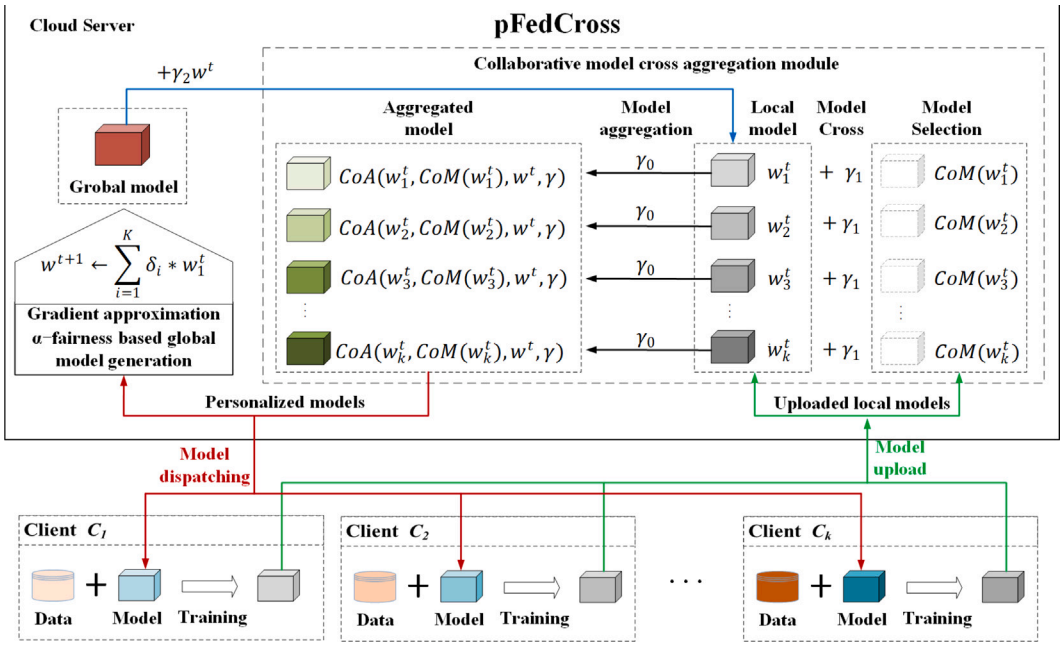

Fig. 1. Framework and workflow of pFedCross.

accuracy deterioration is almost inevitable when dealing with Non-IID or heterogeneous data (Zhu et al., 2021). The primary cause of performance degradation in FL is the weight divergence of local models resulting from Non-IID data. Local models, even with the same initial parameters, will converge to different models due to the heterogeneity in local data distributions. The divergence between the shared global model (the model acquired by averaging the uploaded local models) and the ideal model (the model obtained if the data on local devices were IID) continues to increase. This divergence slows down convergence and worsens learning performance.

To tackle the challenges arising from Non-IID data in FL, researchers have proposed a variety of strategies, These include the creation of a globally shared data subset among edge devices to enhance training on Non-IID data (Zeng et al., 2023), and the development of multi-objective federated learning approaches that balance global performance with individual fairness (Shen et al., 2025). Fairness, in this context, refers to the global model's ability to equitably allocate its performance resources to each client, with a fairer global model being more generalized.

Furthermore, personalized federated learning (PFL) (Sabah et al., 2024) emerges as another potential solution. PFL customizes local models, enabling each model to adapt to its specific local dataset. For instance, FedAMP (Huang et al., 2020) incorporates an attentive message-passing mechanism to assess the importance of other local models in the aggregation of the personalized model. FedALA (Zhang et al., 2023) dynamically updates the aggregate weights of personalized models to generate the global model. MiniPFL (Fan et al., 2024) constructs a bi-component framework that selectively prioritizes valuable clients for optimizing PFL.

We have identified a critical issue where the global gradient may align too closely with the local gradient in the dominant direction, potentially overlooking the rich information contained within the local gradient. This realization underscores the need for more nuanced approaches that address data heterogeneity and ensure the global model remains sensitive to the diverse information present in local gradients.

In this paper, we introduce a novel personalized federated cross-learning framework, pFedCross, aimed at overcoming challenges such as global model performance degradation and the uneven distribution of local models. The framework diagram of pFedCross is depicted in Fig. 1. pFedCross utilizes a complementary collaboration model during the cross aggregation phase, personalizing the local models based on model similarity. Then, the global model is aggregated by addressing the α-fairness through a gradient approximation algorithm, to relieve gradient conflicts.

The main contributions of this paper are summarized as follows:

- We introduce pFedCross, a novel personalized federated cross-learning framework that generates personalized local models with a well-generalized global model.
- We propose a collaborative model cross aggregation algorithm for personalized local models. Utilizing a gradient divergence minimization criterion, we identify a similar collaborative model to minimize gradient divergence. This algorithm adjusts local model parameters to align more closely with the characteristics of local training data, enabling the development of fine-grained local models.
- We present a global model optimization algorithm based on gradient approximation to reach an α-fairness. From a global perspective, the highest similarity strategy for local model updates conflicts with the principle of diversity in participation and knowledge sharing. Therefore, we introduce a gradient based approximation algorithm to reach α-fairness, which is used to adjust the weight based on each local model's accuracy, achieving a balance between fairness and efficiency. This approach resolves conflicting gradients to produce a well-generalized global model.
- The experimental evaluation demonstrates that our proposed approach outperforms the state-of-the-art methods, achieving higher accuracy and lower training loss.

Through these contributions, pFedCross not only enhances the personalization of local models but also optimizes the global model aggregation process, ensuring a more equitable distribution of performance across all participating clients while maintaining high accuracy and efficiency.

This paper is structured as follows: Section 2 provides an overview of related work in FL and PFL addressing issues of performance degradation in Non-IID scenarios, and AI/FL based IDS approaches. Section 3 outlines the architecture of pFedCross, exploring theoretical challenges. Section 4 presents the algorithms of pFedCross. Subsequently, Section 5 offers an evaluative analysis by comparing pFedCross with some typical approaches. The paper concludes in Section 6, summarizing the key findings and suggesting future directions.

Table 1
Summary of existing works on FL-based IDS.

Reference	PFL	Neural network	Main contributions
Rey et al. (2022)	–	MLP Auto-encoder	Evaluates the resilience of FL models against malicious clients
FL-IIDS (Jin et al., 2024)	–	DNN, RNN CNN, GRU	Solves the problem of catastrophic forgetting
VAN-FED-IDS (Chen et al., 2024)	–	Bi-LSTM	Combines Bi-LSTM-based and tree-based models to deliver robust IDS for VANET
IoTDefender (Fan et al., 2020)	✓	CNN	Transfer-learning-based IDS for 5G IoT
pFL-IDS (Thein et al., 2024)	✓	CNN	PFL for poisoned client detect
Ours (pFedCross)	✓	LSTM-Transformer	PFL-based IDS for Non-IID dataset

2. Related work

This section provides the background related to AI-based intrusion detection approaches, FL and PFL addressing issues of performance degradation in Non-IID scenarios, and existing studies on the FL/PFL-based IDS approaches.

2.1. AI-based intrusion detection approaches

ML/DL techniques have shown significant efficacy in detecting anomalies within IoT traffic (Zhang et al., 2025). Mahadevappa et al. (2024) combined linear discriminant analysis (LDA) and logistic regression (LR) for intrusion detection, enabling fast and accurate identification of intrusions. Berguiga et al. (2023) integrated multilayer perceptrons and gaussian mixture models to improve smart agriculture security. Gaber et al. (2023) utilized Kernel principal component analysis for feature extraction and convolutional neural networks (CNNs) for attack classification. Jayalaxmi et al. (2023) developed an autoencoder for feature selection and a cascaded feedforward backpropagation neural network for attack detection in industrial IoT (IIoT) environments. Shoukat et al. (2025) develop an explainable AI (XAI) for multi-class threat classification in IIoT environments. Besides, a variety of DL methods have achieved promising results for IIoT environments, including those leveraging CNN (Wang et al., 2024), recurrent neural networks (RNNs) (Liu and Shi, 2023), long short-term memory networks (LSTMs) (Smith and Wang, 2023), deep Q-Networks (DQN) (Yu et al., 2024b; Shen et al., 2024a), and double DQNP (Yu et al., 2024a; Shen et al., 2024b, 2023, 2024c).

However, these methods typically rely on centralized data analysis, requiring the aggregation of data from all user devices to a single location (edge or cloud server). This approaches not only increases transmission costs but also poses privacy risks, as the data may contain sensitive or confidential information. Besides, above literature rarely considers the impact of Non-IID data on model training accuracy.

2.2. FL algorithms and non-IID data

FL is a collaborative approach where each client trains a local model using its own data and then shares the model parameters with a central server. This exchange occurs in a peer-to-peer fashion among clients via the server, which aggregates the local model parameters to construct a unified global model. This global model is then distributed back to the clients for further refinement through multiple communication rounds, progressively incorporating knowledge from diverse clients and enhancing its accuracy with each iteration.

As a mainstream FL algorithm, FedAvg (McMahan et al., 2017) synthesizes client model weights during the aggregation phase to form the global model after each training round. FedProx (Li et al., 2020b) enhances FedAvg by incorporating constraints on local optimization to restrict the size of local model updates, which helps control model update stability and consistency, especially in cases of non-uniform data distribution or varying client computational capabilities. Similarly, Roth et al. (2021) employed neural architecture search to develop a super-network, consisting of sub-networks tailored for the diverse client datasets.

Despite these innovations, the homogeneous averaging of model parameters can still result in performance degradation, especially in the case of Non-IID data. The uneven distribution feature of Non-IID data introduces biases during model training. Thereby, Wang et al. (2020) proposed a method to alleviate the negative impacts of data heterogeneity by tackling divergence issues stemming from Non-IID datasets. Li et al. (2020a) introduced a committee-based cross-validation approach to refine the global model. Tang et al. (2023) employed a hybrid strategy to enhance the diversity of Non-IID data in training. Hybrid-FL (Yoshida et al., 2020) facilitates centralized data collection from low-privacy-sensitive devices to create IID data for training and global model aggregation. Jin et al. (2024) introduced an aggregation algorithm for model selection, enhancing the global model's accuracy by identifying and excluding biased models in local updates.

2.3. Personalized federated learning (PFL) algorithms for non-IID data

Most FL approaches assume the existence of a single global model that is intended to capture the information from all clients, which is challenging in heterogeneous data environments. Personalized federated learning (PFL) algorithms have been introduced to address this issue (Zhang and Liu, 2023; Hussain et al., 2024). PFL aims to tailor models to meet the unique characteristics of individual clients, providing an effective solution to the accuracy decline experienced by traditional FL algorithms when handling Non-IID data. It effectively reduces data heterogeneity, improving model inference accuracy and enabling more efficient knowledge sharing within a similar convergence time frame.

Hu et al. (2024) proposed FedCross,an efficient FL framework that utilizes a novel multi-to-multi FL training scheme with a multi-model cross aggregation method. FedCross employs a limited number of distinct intermediate models for weighted fusion, to reduce conflicts in convergence direction adjustments. Further, FedCrossEns (Xu et al., 2023) integrates FedCross with ensemble learning to develop federated cross ensemble learning. Extensive efforts have been devoted to developing personalized approaches for Non-IID data using FL algorithms. These efforts include techniques like multi-task learning, model clustering, model parameter decoupling, and knowledge distillation (Rahman et al., 2020).

2.4. FL/PFL-based intrusion detection approaches

FL has risen as an innovative solution in IDS, enabling model training without the need to exchange private data (Zhang et al., 2025). For instance, Rey et al. (2022) proposed a federated learning-based approach for IoT network anomaly detection that combines supervised deep learning and unsupervised auto-encoder techniques. VAN-IDS (Chen et al., 2024) integrates Bi-LSTM for packet analysis and LightGBM for physical feature analysis. IoTDefender (Fan et al., 2020) utilized federated transfer learning to address security challenges

in 5G-enabled IoT environments. pFL-IDS (Thein et al., 2024) manages imbalanced data distributions and employed a mini-batch logit adjustment loss for personalized local model training. These approaches highlight the potential of FL in enhancing security while maintaining data privacy in IoT environments.

As listed in Table 1, the aforementioned studies demonstrate the effectiveness of FL/PFL-based IDS. However, most of them overlook the impact of data heterogeneity on model convergence and differentiation of the local model for heterogeneous data. Addressing these issues is essential for improving the performance of FL-based IDS. Consequently, we propose a novel solutions, pFedCross to customize the personalized fine-grained local models while producing a well-generalized global model without lossing model convergence and stability.

In summary, while various approaches, such as FedAvg, FedProx, and FedCross, have contributed to the field by addressing challenges like data heterogeneity and privacy concerns, our work presents a novel framework, pFedCross, which introduces a personalized federated cross-learning strategy to further refine the training of local models and the aggregation of a global model. In comparison to existing methods, pFedCross stands out by effectively managing Non-IID data through a collaborative model cross-aggregation algorithm, which minimizes gradient divergence, and by dynamically adjusting local model parameters to align with local training data characteristics. This innovative ensures that the global model benefits from the diversity of local datasets without compromising on the convergence and stability of the training process.

3. System architecture

3.1. System overview

As illustrated in Fig. 1, the architecture of pFedCross framework is composed of a central cloud server and K distributed participants, or clients, denoted as C_i, $i = \{1, \dots, K\}$. Each client C_i manages a private dataset $D_i = \{x_i, y_i\}$, with x_i denoting the data sample and y_i representing their corresponding label.

Clients in the system collect and store data from a variety of IoT devices, performing essential preliminary processing steps. These include data cleaning to ensure accuracy, format conversion for consistency, and data compression to optimize storage and transmission efficiency. Such pre-processing activities are vital for streamlining subsequent data analysis and model training procedures.

The cloud server plays a pivotal role by aggregating the local models from various clients, synthesizing a global model that combining the collective insights of the distributed clients. It is responsible for distributing the personalized local models back to the clients, facilitating further training and refinement. These localized models align with the unique characteristics of each client's data, which is instrumental in effectively capturing localized patterns and nuances.

The components of pFedCross are elaborated as follows.

- **Global model initialization**: The cloud server initiates the process by initializing a global model, which is then distributed to the participating clients in the first round.
- **Local training**: Clients proceed to train the received model with their private datasets, subsequently updating their local models, uploading both the updated local models and their respective accuracy to the cloud server.
- **Collaborative model selection**: To improve model convergence, the collaborative model is selected based on the highest similarity among clients' models. Model similarity is assessed using a weighted Euclidean distance method.
- **Model aggregation**: Personalized local models are refined by aggregating the local model, the collaborative model, and the global model with predefined weights, creating a more nuanced model for each client.
- **Global model update**: Acknowledging the variability of local models, we incorporate a gradient approximation algorithm designed to achieve α-fairness, ensuring that the global model training process accounts for local model diversity, thereby improving overall performance and robustness. The global model is then regenerated based on the weighted contributions of local models.
- **Model dispatching**: The cloud server subsequently dispatches the updated personalized models back to the clients, where it can be further trained and refined, continuing the cycle of FL.

3.2. Theoretical challenges of the non-IID data in FL

The objective of FL is to minimize the global loss function $L(w)$ while collaboratively train a global model utilizing the isolated datasets $\{D_0, \dots, D_K\}$. Let w_i^t represent the model parameters for client i in tth round, N_i denote the total number of data samples in D_i, and $\hat{y}_i$ be the predicted probability that a sample belongs to the positive class. The total number of samples across all datasets is represented by D. $|D_i|$ is the size of client i's dataset. The loss function for the dataset D_i is denoted by $L(w_i)$. In this paper, the cross-entropy loss is used for model optimization, as defined in (3). The objective can be articulated by:

$$\min L(w) = \min \sum_{i=1}^{K} \frac{|D_i|}{D} L(w_i), \tag{1}$$

$$D = \sum_{i=1}^{K} |D_i|, \tag{2}$$

$$L(w_i) = -\frac{1}{N_i} \sum_{i=1}^{N_i} \left[y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i) \right]. \tag{3}$$

During each round of iteration, client i employs gradient descent technique to update the parameters of local model. Specifically, the parameter update for client i in the (t+1)th round is given by:

$$w_i^{t+1} = w_i^t - \lambda_i^t \nabla L(w_i^t), \tag{4}$$

where, $\nabla L(w_i^t)$ is the average gradient of dataset D_i in tth round, and λ_i^t is the step size, or learning rate, for client i at that round.

Subsequently, the global parameter w^{t+1} is updated using the following aggregation formula:

$$w^{t+1} = \sum_{i=1}^{K} \frac{|D_i|}{D} w_i^{t+1}. \tag{5}$$

By combining Eqs. (4) and (5), we derive the expression for the updated global model parameters:

$$w^{t+1} = w^0 - \sum_{i=1}^{K} \frac{|D_i|}{D} \lambda_i^1 \nabla L(w_i^0) - \sum_{i=1}^{K} \frac{|D_i|}{D} \sum_{t=2}^{T} \lambda_i^t \nabla L(w_i^t). \tag{6}$$

Examining (6), the second term on the right-side is essentially the gradient of the global objective function, which is a descent direction of the global objective function $L(w)$. $\nabla L(w_i^0)$ represents the initial gradient of the loss function for client i's dataset. However, the third term may not consistently point in a descent direction due to discrepancies among w_i^t across different clients C_i. When the datasets are Non-IID, these discrepancies can grow significantly, making it challenging to evaluate the quality of the updated model without additional assumptions. This can ultimately lead to the degradation of the global model's performance, as the divergence between the shared global model and the ideal model continues to increase, impeding convergence and diminishing learning efficacy.

3.3. Summary of main notations and abbreviations

The main notations and abbreviations are summarized in Table 2.

Table 2
Summary of main notations and abbreviations.

Notations	Definition
C_i	The ith client in K participants, $i = \{1, \dots, K\}$
D_i	The dataset of the client i
$\lvert D_i \rvert$	The size of the dataset D_i
K	Number of clients
x_j	The jth data point
y_j	The jth corresponding label
w_i	Model parameters for client i
w	Global model parameters
W	The collection of local model parameters
$L(w)$	The global loss function
$L(w_i^t)$	The loss function of client i in tth iteration
$\nabla L(w_i^t)$	The gradient of dataset D_i in tth round
α	Fairness parameter
β, γ, δ	Weights of models or model parameters
λ	The learning rate or step size
$E(\alpha)$	The system efficiency function
$F(\alpha)$	The system fairness function
ε	Validity measure parameter
δ	Optimal weights for local models

4. Algorithms of personalized federated cross-learning (pFedCross)

4.1. Preliminaries

Many FL algorithms enable local models to independently minimize the loss on their respective datasets. However, the aggregated model may not always reach the global minimum, particularly in scenarios with Non-IID datasets, where divergence among local models can increase rapidly with each communication round. To address this challenge, we introduce pFedCross, a framework that employs personalized federated cross-learning to train multiple models concurrently and deploy them in parallel.

In pFedCross, the cloud server strategically selects complementary collaborative models and aggregates them by weight to form a personalized local model. The collaborative model can be selected according to the highest similarity strategy to reach the lowest gradient divergence. This approach makes clients with similar data attributes more aggregated with faster convergence, but impacts the generality of the global model. Therefore, the global model is optimized based on gradient approximation algorithm to reach α-fairness. This approach aims to harmonize the strengths of diverse local models while countering the challenges posed by Non-IID datasets, leading to a more robust and accurate global model.

The primary goal of pFedCross is to enhance both the generalization capability and training efficiency of the global model through optimized client selection and refined local training processes, ultimately boosting the classification performance and efficiency of these personalized models in heterogeneous environments. The comprehensive algorithms that collectively form the pFedCross framework, as depicted in Fig. 1, are further elaborated in the subsequent sections.

4.2. Model cross aggregation algorithm for personalized local models

Selecting the collaborative model is a pivotal step in pFedCross. Generally, there are three fundamental criteria for model selection: (1) ensuring adequacy and diversity among participating models, (2) minimizing the differences in gradients, and (3) maximizing the acquisition of new knowledge. Given that models trained across various clients naturally exhibit differences, our approach emphasizes the minimization of gradient divergence. This criterion prompts each model to identify a collaborative model that is similar, thereby reducing gradient divergence during each aggregation step. Based on this criterion, we present the highest similarity strategy:

$$w_j^t \leftarrow CoM(w_i^t, W) = \arg\max_{w_j^t \in W} Similarity(w_i^t, w_j^t), \tag{7}$$

Algorithm 1 Collaborative model selection and aggregation algorithm

Input: local model parameter set W, parameters $\{\beta_1, \beta_2, \dots, \beta_n\}$, parameters $\{\gamma_1, \gamma_2, \gamma_3\}$, initial global model w^0, number of clients K, number of iterations T
Output: W

$W \leftarrow [w_1^0, w_2^0, \dots, w_k^0]$
for $t = 0, \dots, T-1$ **do**
  for $i = 0, \dots, K-1$ **do**
    $w_j^t \leftarrow CoM(w_i^t, W)$
    $w_i^{t+1} \leftarrow \underset{k \in \{0,1,2\}}{CoA}(w_i^t, w_j^t, w^t, \gamma_k)$
    $t = t + 1$
  end for
end for
$W \leftarrow [w_1^t, w_2^t, \dots, w_k^t]$
Return W

where W represents the list of uploaded local model parameters.

Model similarity in the proposed framework is determined using a weighted Eucledian distance metric, which quantifies the spatial distance between two vectors. We account for the varying importance of model parameters by integrating a weight factor β into the Euclidean distance formula, as depicted below:

$$D_w^t = \sqrt{\sum_{i=1}^{n} \beta_i^t \cdot (X_i^t - Y_i^t)^2}, \tag{8}$$

where X_i and Y_i denote the ith parameters of models X and Y, respectively, and n is the total count of parameters. The similarity between two models is formulated as:

$$Similarity(w_i^t, w_j^t) = 1/(1 + D_w^t). \tag{9}$$

Upon the identification of the collaborative model, model aggregation proceeds to derive the updated local model parameters w_i^{t+1} for client i, as elaborated in (10). The updated local model is the product of aggregating the parameter of the local model w_i^t, the collaborative mode w_j^t, and the global model w^t. We introduce weights γ_i ($i \in \{0, 1, 2\}$) ($0 \le \gamma \le 1$) to determine the importance of each model in the aggregation process. The impact of varying γ values on the performance of pFedCross is further investigated in Section 5.4.2.

$$w_i^{t+1} \leftarrow \underset{k \in \{1,2,3\}}{CoA}(w_i^t, w_j^t, w^t, \gamma_k) = \gamma_0 * w_i^t + \gamma_1 * w_j^t + \gamma_2 * w^t, \tag{10}$$

$$\text{s.t.} \quad \sum_{i=0}^{2} \gamma_i = 1. \tag{11}$$

The collaborative model selection and aggregation algorithm is expressed by the pseudo-code in Algorithm 1.

4.3. Global model optimization based on gradient approximation algorithm

The highest similarity strategy minimizes gradient divergence in local model aggregation but can lead to a lack of diversity in the global model, as similar models become more alike and dissimilar models struggle to share knowledge. This can result in distinct optimization directions among local model groups by the end of FL training, causing significant gradient conflicts during global model generation. To enhance the generalization of global models, it is crucial to balance fairness and diversity in the face of heterogeneous datasets.

In the pFedCross framework, to balance fairness and efficiency, we propose a gradient approximation algorithm based on weighted α-fairness. This algorithm adjusts the weight δ according to the accuracy

Algorithm 2 Gradient approximation algorithm to achieve α-fairness

Input: Accuracy of local models $\{A_0, A_1, \ldots, A_{K-1}\}$, validity threshold E_T, tolerance error e, step size λ, positive real number ϵ
Output: Fairness parameter α

Initialization: i=0
while $|E(\alpha_i) - E_T| \geq e$ **do**
  $\alpha_{i+1} \leftarrow (\alpha_i - \frac{E(\alpha_i)-E_T}{E'(\alpha_i)} \cdot \lambda) > 0?(\alpha_i - \frac{E(\alpha_i)-E_T}{E'(\alpha_i)} \cdot \lambda) : \epsilon$
  if $(E(\alpha_{i+1}) - E_T) \cdot (E(\alpha_i) - E_T) < 0$ **then**
    $\lambda = \frac{\lambda}{2}$
  end if
  $i = i + 1$
end while
Return α_{i+1}

Algorithm 3 Global model update process

Input: Local model parameters W from Algorithm 1, Fairness parameter α from Algorithm 2
Output: Global model w

Initialization: w^0, $\{\delta_0, \ldots, \delta_K\}$
$W \leftarrow [w_1^t, w_2^t, \ldots, w_k^t]$
for $t = 0, \ldots, T-1$ **do**
  for $i = 0, \ldots, K-1$ **do**
    $A_i^t \leftarrow Clientaccuary(D_i, w_i^t)$
    $\delta_j = A_j^{\frac{1-\alpha}{\alpha}} / \sum_{i=1}^{K} A_i^{\frac{1-\alpha}{\alpha}}$
  end for
end for
$w^{t+1} \leftarrow \sum_{i=1}^{K} \delta_i * w_i^t$
Return w^{t+1}

A of each local model. Specifically, for client j with model accuracy A_j, its weight δ can be expressed as:

$$\delta_j = A_j^{\frac{1-\alpha}{\alpha}} / \sum_{i=1}^{K} A_i^{\frac{1-\alpha}{\alpha}}. \tag{12}$$

Let $E(\alpha)$ and $F(\alpha)$ be the system efficiency and system fairness, respectively. Under α-fairness (Guo et al., 2013), the system efficiency is given by:

$$E(\alpha) = \sum_{i=1}^{K} A_i^{\frac{1}{\alpha}} \Big/ \sum_{i=1}^{K} A_i^{\frac{1-\alpha}{\alpha}}. \tag{13}$$

The system fairness can be formulated as:

$$F(\alpha) = (\sum_{i=1}^{K} A_i^{\frac{1}{\alpha}})^2 \Big/ K \sum_{i=1}^{K} A_i^{\frac{2}{\alpha}}. \tag{14}$$

According to the definition of (13), the differential of $E(\alpha)$ is:

$$E'(\alpha) = \frac{E(\alpha_{i+1}) - E(\alpha_i)}{\alpha_{i+1} - \alpha_i}. \tag{15}$$

A larger value of $E(\alpha)$ indicates that the aggregation model is more effective with a higher weighted average accuracy. A larger value of $F(\alpha)$ indicates that the aggregated model is fairer, with accuracy more evenly distributed among participants.

To find the optimal α, the objective is to maximize the fairness while ensuring that the efficiency threshold E_T is met. This can be formulated as the following optimization problem:

$$\max_{\alpha} F(\alpha) \quad \text{s.t.} \quad E(\alpha) \geq E_T, \tag{16}$$

The algorithm is summarized in Algorithm 2 and outlined below. In the algorithm, the parameters are defined as follows: i is the iteration index, e is the tolerance error, α_i is the value of α after the ith iteration, λ is the step size factor, ϵ is a small positive real number used to adjust the step size. The condition $(E(\alpha_{i+1}) - E_T) \cdot (E(\alpha_i) - E_T) < 0$ checks if the change in α_i results in crossing the threshold E_T, which would require reducing the step size to fine-tune the parameter more accurately. This gradient descent approach effectively navigates the trade-off between efficiency and fairness, iteratively adjusting α to find the optimal value that maximizes system fairness while maintaining the required efficiency threshold.

Once the optimal parameter α has been determined, the weight δ for each client can be calculated using the previously defined formula (12). With the weights δ_i, the global model can be updated by:

$$w^{t+1} \leftarrow \sum_{i=1}^{K} \delta_i * w_i^t. \tag{17}$$

Algorithm 3 outlines the process of aggregating the global model based on a gradient approximation algorithm, which is designed to achieve α-fairness by accounting for disparities among local models. This method contrasts with traditional approaches that simply average local model parameters. The initial inputs are the local model parameters W from Algorithm 1 and fairness parameter α from Algorithm 2. The model weights are calculated using (12), which is then applied in (17) to derive the updated global model.

In this way, we can mitigate the impact of Non-IID data on model aggregation while ensuring that the contribution of different clients to the global model is aligned with their model performance. This weighted α-fairness strategy helps us better handle data heterogeneity and improve the robustness and accuracy of the global model. By dynamically adjusting the weights of each client's model, we can maintain model diversity while optimizing the performance of the global model. This approach is not only applicable to the pFedCross framework but can also be extended to other federated learning scenarios where model performance and data fairness need to be considered.

5. Experiments and evaluation

We evaluated the performance of our pFedCross with two typical datasets and compared it with several state-of-the-art approaches to investigate the effectiveness of the proposed method. The detail of the dataset, experiment setting, and evaluations are provided in this section.

5.1. Experimental settings

Experimental environment: Our experiments were conducted with the PyTorch framework and executed on a high-performance server. The server was equipped with an Intel Xeon(R) Silver 4216 processor (2.1 GHz*16), 64 GB of DRAM, and 8 TB of storage to handle the computational demands of our FL and deep learning experiments.

Dataset: We evaluated our method using four publicly available datasets: the NSL-KDD dataset (Ban et al., 2024), the CICIDS2017 dataset (Wen et al., 2025), the CICIoT2023 dataset (Behera et al., 2024), and the BoT-IoT dataset (Luqman et al., 2025). The NSL-KDD dataset, a widely used benchmark for intrusion detection, improves upon the KDDCup99 dataset. It includes 148,517 records with 41 features and labels for 22 types of cyberattacks. The CICIDS2017 dataset provides network traffic data generated by CICFlowMeter, capturing flow details such as timestamps, IP addresses, ports, protocols, and attack types. The CICIoT2023 dataset is a recent and comprehensive collection specifically designed to validate security analysis applications in IoT environments. It contains 2.6 million samples with 47 features, categorized into 7 distinct classes. The BoT-IoT dataset consists of over 70 million samples with 46 features. For efficiency, we used a 5% subset that retains all features while reducing the data volume.

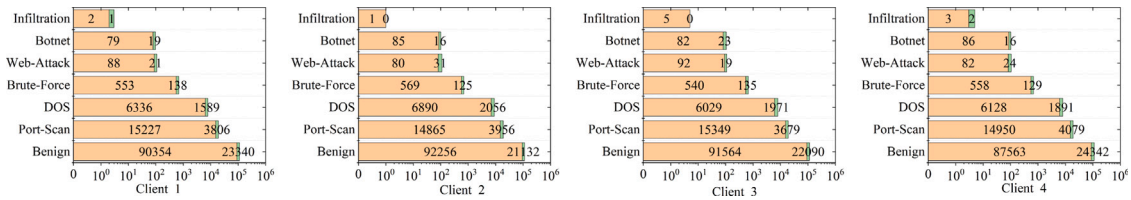

Fig. 2. Label distribution in IID scenario (train, test).

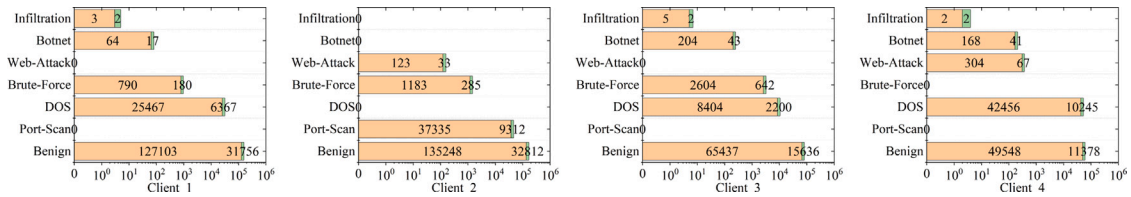

Fig. 3. Label distribution in Non-IID scenario (train, test).

Table 3
The BoT-IoT dataset.

Class label	Number of instances
Normal	9543
DoS	1,650,260
DDoS	1,926,624
Reconnaissance	91,082
Theft	79

Table 4
The CICIDS2017 dataset.

Class label	Number of instances
Normal	2,273,097
Port-Scan	380,699
DoS	158,930
Brute-Force	13,835
Web-Attack	2180
Botnet	1966
Infiltration	36

Due to the scarcity of normal data samples in this subset, all normal data from the BoT-IoT dataset were added to this subset to better facilitate subsequent experiments. Take the BoT-IoT dataset and the CICIDS2017 dataset as the example, the detailed traffic classifications are provided in Tables 3 and 4 respectively.

Data distribution: In the simulation, the dataset is partitioned among 20 clients to mimic real-world data distribution. We simulate both IID (Independent and Identically Distributed) and Non-IID data scenarios across the clients. For instance, as depicted in Fig. 2, the CICIDS2017 dataset is randomly and uniformly distributed among the clients, with each client exhibiting similar traffic characteristics. Consequently, all clients are subjected to the same frequency of attacks, which represents an IID data scenario. In contrast, as illustrated in Fig. 3, the Non-IID scenario, which is more realistic, challenges the assumption that all clients experience the same types of attacks. In this scenario, attacks are not evenly distributed across all clients; certain attacks are only present on specific clients, thus accurately simulating a Non-IID scenario. Other data sets that simulate IID (independent and equally distributed) and non-IID data scenarios on the client side are treated similarly to the CICIDS2017 dataset.

Data preprocessing: The Min-Max Scaler (18) is used to normalize the initial indicators of the eigenvector matrices into the range of $[0, 1]$. Here, $X_{\min}$ and $X_{\max}$ denote the minimum and maximum values of the features, respectively, while the normalized feature value is represented by X_n, $0 < X_n < 1$.

$$X_n = \frac{X - X_{\min}}{X_{\max} - X_{\min}}. \tag{18}$$

Table 5
Confusion matrix.

	Predict as "Attack"	Predict as "Normal"
Sample is "Attack"	True Positive (TP)	False Negative (FN)
Sample is "Normal"	False Positive (FP)	True Negative (TN)

Evaluation metrics: In our experiments, "attack" is defined as the positive class. As defined in Table 5, a false positive (FP) occurs when the model incorrectly classifies a sample from normal traffic as an attack, while false negative (FN) incorrectly classifies attack sample as normal. Conversely, a true negative (TN) and true positive (TP) denote the number of normal or attack data that were accurately classified.

Based on these values, the evaluation metrics are computed, including Accuracy (ACC), Positive Predictive Value (PPV), True Positive Rate (TPR), and F1 Score. ACC is the proportion of total samples that were correctly classified. PPV is the ratio of TP results to all positive results (TP+FP). TPR is the ratio of TP results to all actual positive cases (TP+FN). F1 score is the harmonic mean of precision and recall (19), providing a single measure that balances both metrics, which is particularly useful for datasets with imbalanced classes. F1 score is especially useful for skewed datasets as it offers a balanced measure of precision and recall, crucial for models where high accuracy and correct identification of positive instances are paramount.

$$\mathrm{F1} = 2 \cdot \frac{PPV \cdot TPR}{PPV + TPR}. \tag{19}$$

5.2. Feature extraction based on LSTM and transformer

In this paper, we use a hybrid model that combines LSTM networks and Transformer architecture (Ullah et al., 2023) to form the foundation of local model. The LSTM is particularly adept at extracting time series features from network traffic, while the Transformer efficiently learns spatial features.

In order to capture the evolution of dynamic trends and pattern characteristics in traffic data, a sliding window mechanism is applied in conjunction with LSTM. As the window moves forward, the model continually receives and analyzes new segments of traffic data, facilitating real-time model training. The Transformer then integrates and compresses the features extracted by LSTM at a deeper level, performing dimension reduction and secondary feature extraction to uncover intrinsic changes in the data. The output of the Transformer layer, denoted as X_{out}, is given by:

$$X_{out} = \partial_{comb} X_{in} + \zeta_p MLP(Norm(X_{in})) + \zeta_a MHA(Norm(X_{in})), \tag{20}$$

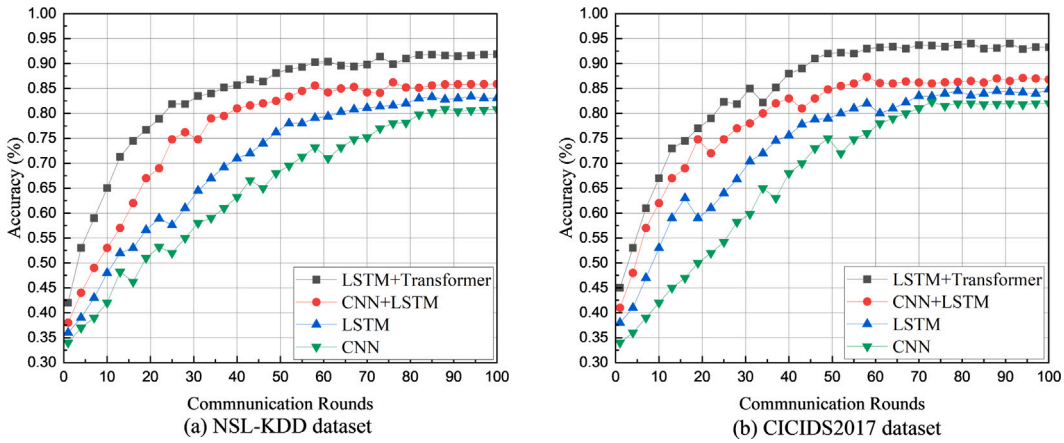

Fig. 4. Convergence comparison of different models.

where ∂_{comb} represents the scale factor of the input contribution, ζ_p and ζ_a are the trainable parameters for the multi-layer perceptron (MLP) and multi-head attention (MHA) blocks, respectively. The normalization function, denoted as Norm, is applied to the input before it is processed by the MLP and MHA blocks.

Following feature extraction, three components of network traffic representation are identified: the hidden representation, reconstruction residual vector, and reconstruction error. These components highlight deviations of abnormal samples from the normal pattern. An anomaly scoring module assesses the normalcy of features generated by the LSTM and Transformer based on these scores. Both clients and the server compute losses from these scores, updating the model until convergence. An MLP is employed in this study for anomaly scoring:

$$S_i = Sigmoid(f_{MLP}(h_i)), \tag{21}$$

where f_{MLP} is the MLP on the server, h_i is the feature generated by the feature representation of the ith device, and S_i is the anomaly score. A higher anomaly score indicates a higher probability that the data is abnormal, and a lower anomaly score indicates a higher probability that the data is normal.

5.3. Baseline algorithms

In our experiments for one-model FL framework, we compare several algorithms, including FedAvg (McMahan et al., 2017), FedProx (Li et al., 2020b), and FedCross (Hu et al., 2024). These algorithms represent different strategies to address the challenges of FL, particularly in terms of handling Non-IID data and improving the overall performance of the trained models. FedAvg is the most classical FL framework, which dispatches a global model to multiple clients for FL training and aggregates the multiple local models averagely to update the global model. FedProx was designed to address the challenges posed by model performance degradation in Non-IID scenarios. FedCross presented a multi-model cross aggregation method to satisfy convergence characteristics of individual clients without averaging their local models.

5.4. Experimental comparison

In our experiments, we conduct performance assessment against established benchmarks (i.e., FegAvg, FedProx, FedCross), as well as a comparative evaluation of various deep learning models (i.e., CNN, LSTM, CNN-LSTM, LSTM-Transformer). The subsequent sections of our study are structured to address the efficacy of these diverse deep learning models, beginning with a validation of our chosen model. Subsequently, we exam the accuracy and convergence of the model across different FL frameworks. Finally, the model training loss and complexity analysis are presented.

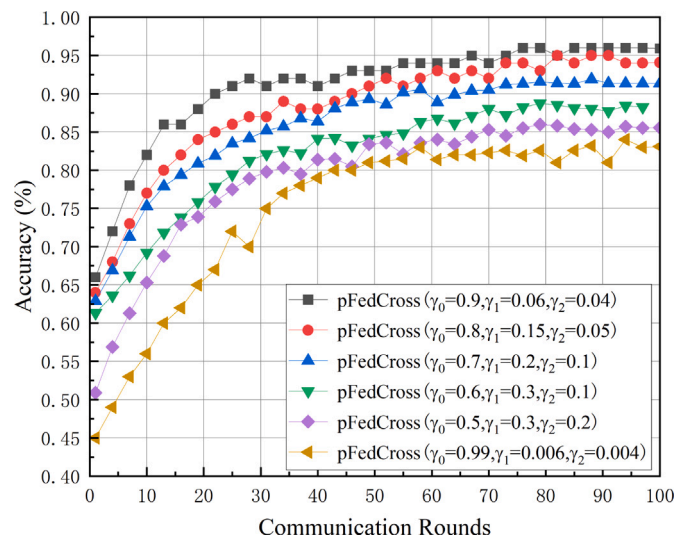

Fig. 5. The impact of aggregation rate on model accuracy.

5.4.1. Performance comparison of different models

Fig. 4 shows the experimental results of different deep learning models on the NSL-KDD dataset and CICIDS2017 dataset, respectively. It is evident that the LSTM-Transformer model outperforms the others in terms of accuracy. Regarding model convergence, both the LSTM-Transformer and CNN-LSTM models achieve stable convergence after approximately 60 communication rounds across both datasets. In comparison, the CNN and LSTM models exhibit a more gradual convergence, stabilizing around 70 iterations. Consequently, the client of pFedCross in this study utilizes the LSTM-Transformer as the local model for comparative training.

5.4.2. Evaluation of aggregation rate

Eq. (10) and the discussion in Section 4.2 highlight the critical influence of the aggregation rate on the weighting of the previous local model, the collaborative model, and the global model when crafting personalized local models for further training. The weights γ_1, γ_2, γ_3 are dynamically adjusted to assess the aggregation rate's impact on model accuracy. The experimental results, depicted in Fig. 5, demonstrate the application of the LSTM-Transformer model on pFedCross framework.

It is evident that the performance of pFedCross gradually declines as the value of γ_1 decreases. Notably, there is a significant drop in accuracy when γ_1 is set to 0.99. This decline is attributed to the fact that an excessively high value of γ_1 restricts the knowledge gained from both

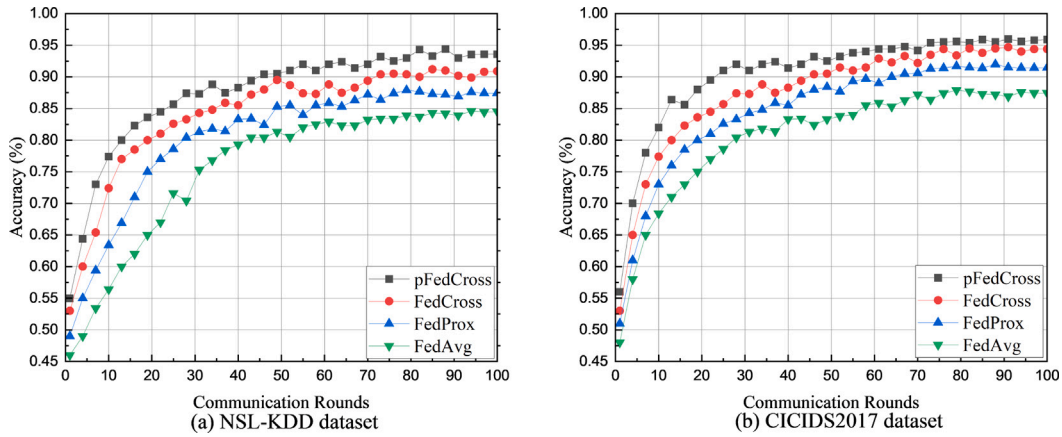

Fig. 6. Convergence of LSTM-Transformer on different FL frameworks.

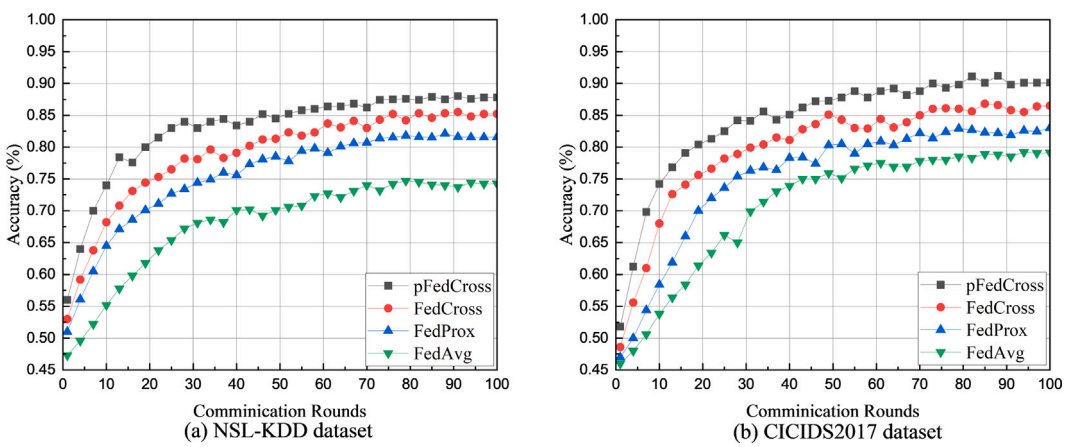

Fig. 7. Convergence of CNN-LSTM on different FL frameworks.

the collaborative model and the global model. Consequently, we set the parameters $\{\gamma_1, \gamma_2, \gamma_3\}$ to {0.9, 0.06, 0.04} in this study to achieve a balance between knowledge acquisition and model distance management. More tests will be carried out to find the optimal combination of $\{\gamma_1, \gamma_2, \gamma_3\}$.

5.4.3. Comparison with baseline frameworks

In this paper, we have employed the Adam optimizer with a learning rate set to $\lambda = 0.001$ and a dropout value of 0.1 to fine-tune our proposed model. Within the context of intrusion detection, where data heterogeneity is a common challenge, our model has demonstrated superior accuracy through comparative analysis with the FedAvg, FedProx, and FedCross algorithms. The experiments were conducted using the LSTM-Transformer model on both the NSL-KDD dataset and the CICIDS2017 dataset. As shown in Table 6, pFedCross outperforms its counterparts in terms of ACC, PPV, TPR, and F1 score, highlighting its effectiveness in intrusion detection scenarios.

5.4.4. Comparison of convergence on different FL frameworks

Fig. 6 demonstrates the convergence trends of the LSTM-Trans former model across various FL methods on both the NSL-KDD dataset and the CICIDS2017 dataset. Our findings indicate that pFedCross consistently achieves the highest accuracy among the four FL methods tested. Moreover, pFedCross exhibits significantly smaller fluctuations in convergence for all models and data settings examined. This stability is primarily attributed to the model cross-aggregation algorithm employed by pFedCross, which facilitates the nuanced training of personalized models. By mitigating gradient divergence during local training, pFedCross not only attains the highest test accuracy but also reduces the probability of becoming ensnared in local minima.

Table 6
Comparison with baselines on dataset NSL-KDD and CICIDS2017.

Dataset	Framework	ACC (%)	PPV (%)	TPR (%)	F1 (%)
NSL-KDD	FedAvg	84.34	85.26	84.78	85.02
	FedProx	87.22	88.79	86.16	87.46
	FedCross	90.49	92.39	89.34	90.84
	pFedCross	**93.59**	**93.49**	**93.19**	**93.34**
CICIDS2017	FedAvg	87.22	88.79	86.16	87.46
	FedProx	92.44	92.13	91.57	91.85
	FedCross	94.57	93.87	92.68	93.27
	pFedCross	**95.91**	**95.29**	**95.83**	**95.56**

In contrast, the CNN-LSTM model, when tested on the same set of FL frameworks. The convergence trends of CNN-LSTM model on FL frameworks are similar with LSTM-Transformer. However, CNN-LSTM model performs less optimally in terms of both accuracy and convergence, as depicted in Fig. 7.

5.4.5. Comparison on model training loss

This study evaluates the training loss of the LSTM-Transformer model on the NSL-KDD dataset and CICIDS2017 dataset, separately. The loss trends are graphically represented in Fig. 8, highlighting the comparative performance of the pFedCross model against other methods. The results indicate that the pFedCross model not only converges more swiftly but also achieves a lower training loss compared to its counterparts.

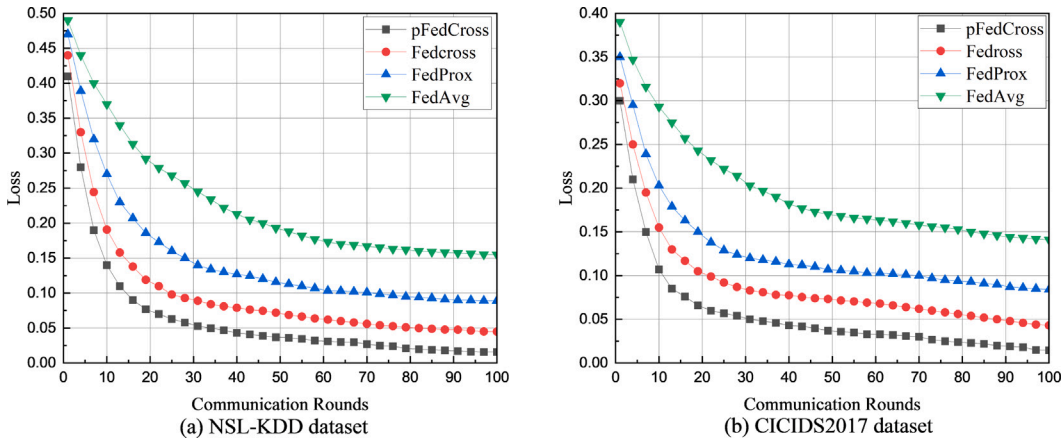

Fig. 8. Comparison of model training loss.

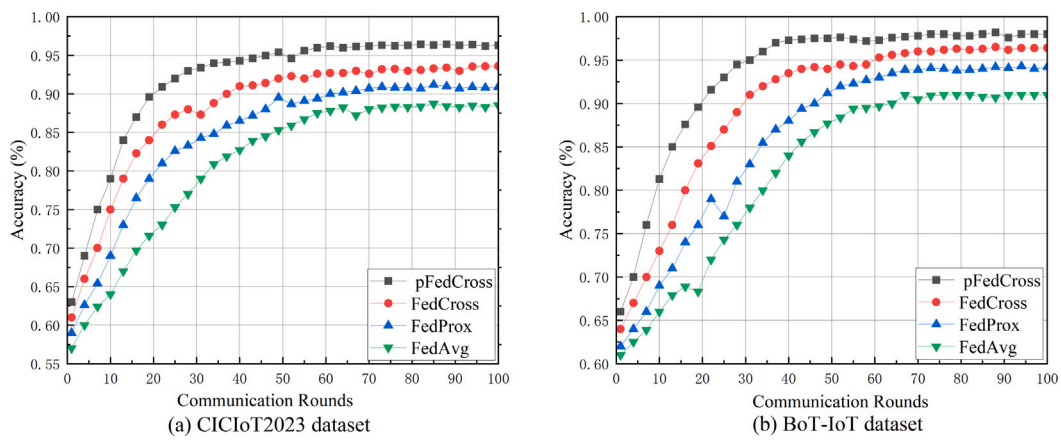

Fig. 9. Convergence of LSTM-Transformer on different FL frameworks.

In the case of the NSL-KDD dataset, as illustrated in Fig. 8(a), the pFedCross model converges to a minimal loss value of 0.021 after 100 training rounds. This represents a significant improvement, translating to an approximately eightfold reduction compared to the FedAvg scheme, which records a loss of 0.1552 under the same conditions. The FedProx and FedCross algorithms report loss values of 0.0892 and 0.0451, respectively, after the same number of training rounds. This data substantiates the superior performance of the pFedCross model, particularly under Non-IID conditions.

For the CICIDS2017 dataset, Fig. 8(b) shows that pFedCross achieves an optimal loss value of 0.015. These findings underscore the robustness and efficacy of the pFedCross model in minimizing training loss across different datasets, thereby highlighting its potential in federated learning scenarios where data heterogeneity is prevalent.

5.5. Performance validation on the CICIoT2023 and BoT-IoT datasets

To rigorously evaluate the efficacy of the proposed model in IoT environments, the LSTM-Transformer architecture was employed for comprehensive experimentation on the CICIoT2023 and BoT-IoT datasets. The performance of pFedCross was systematically assessed, with particular emphasis on the variations in accuracy and loss functions, as illustrated in Figs. 9 and 10.

Fig. 9 presents the accuracy convergence trends of various FL frameworks utilizing the LSTM-Transformer model on the CICIoT2023 and BoT-IoT datasets. The experimental findings reveal that pFedCross demonstrates comparable convergence trends across both datasets. Notably, among the four FL frameworks evaluated, pFedCross exhibits markedly reduced convergence fluctuations and consistently attains superior accuracy.

The training loss trends of the LSTM-Transformer model on the CICIoT2023 and BoT-IoT datasets were meticulously analyzed and are depicted in Fig. 10. The loss function variations on both the CICIoT2023 and BoT-IoT datasets exhibit trends analogous to those observed on the NSL-KDD and CICIDS2017 datasets. In comparison to FedAvg, FedProx, and FedCross, pFedCross demonstrates the most rapid decline in loss values and achieves the minimal loss, underscoring its enhanced performance and stability.

To verify the effectiveness of the pFedCross model in IoT environments, experiments were conducted using the LSTM-Transformer model on the CICIoT2023 and BoT-IoT datasets. The experimental results, as shown in Table 7, indicate that pFedCross outperforms other federated learning algorithms in metrics such as ACC, PPV, TPR, and F1 score. The results demonstrate that pFedCross is highly effective in IoT scenarios.

5.6. Complexity analysis

In this section, we present the complexity analysis to assess the efficiency of the pFedCross framework and the deployed models. We begin by examining the complexity of the FedAvg, FedProx, FedCross, and pFedCross frameworks. FedAvg serves as the foundational method with the lowest complexity due to its simplicity. It involves averaging the model updates from all clients, which is straightforward but may not be optimal in scenarios with stragglers or data heterogeneity. FedProx introduces additional complexity by incorporating regularization terms to mitigate the effects of stragglers and data heterogeneity,

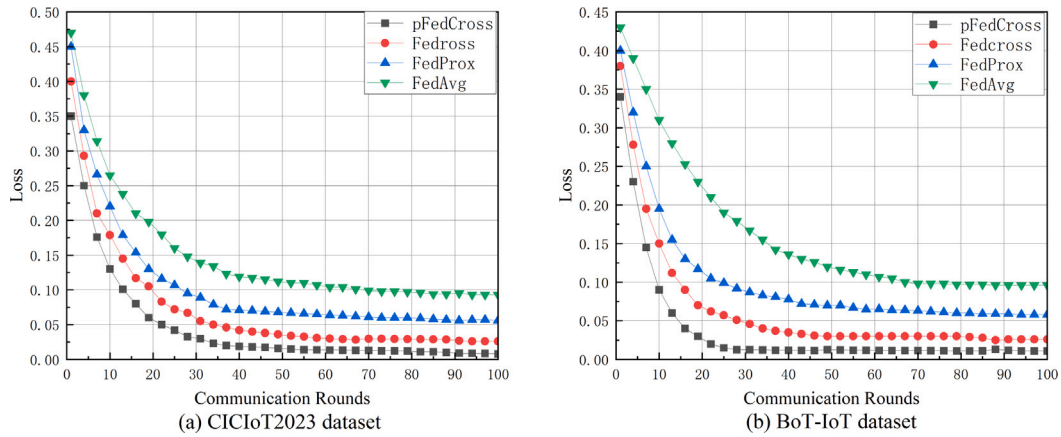

Fig. 10. Comparison of model training loss.

Table 7
Comparison with baselines on CICIoT2023 and BoT-IoT datasets.

Dataset	Framework	ACC (%)	PPV (%)	TPR (%)	F1 (%)
CICIoT2023	FedAvg	88.51	87.18	89.73	85.44
	FedProx	90.91	89.79	90.13	89.96
	FedCross	93.59	92.56	94.21	93.38
	pFedCross	**96.32**	**95.28**	**97.47**	**96.36**
BoT-IoT	FedAvg	91.02	90.29	92.16	91.22
	FedProx	94.24	93.31	95.72	94.50
	FedCross	96.41	94.67	98.18	96.39
	pFedCross	**98.05**	**97.55**	**98.67**	**98.11**

Table 8
Time and space complexity comparison for different models.

Model	Time complexity	Space complexity
CNN	$O(n \cdot k^2 \cdot c_{\text{in}} \cdot c_{\text{out}})$	$O(k^2 \cdot c_{\text{in}} \cdot c_{\text{out}})$
LSTM	$O(n \cdot h^2)$	$O(4 \cdot (h^2 + h \cdot d))$
CNN-LSTM	$O(n \cdot k^2 \cdot c_{\text{in}} \cdot c_{\text{out}} + n \cdot h^2)$	$O(k^2 \cdot c_{\text{in}} \cdot c_{\text{out}} + 4 \cdot (h^2 + h \cdot d))$
LSTM-Transformer	$O(n^2 \cdot d + n \cdot h^2)$	$O(n^2 \cdot d + 4 \cdot (h^2 + h \cdot d))$

thus increasing its computational demands. FedCross and pFedCross take a more elaborated approach by leveraging multiple models for aggregation, which enhances accuracy in Non-IID scenarios. Notably, these methods do not incur additional communication overhead despite their complexity. The communication overhead for all these methods is characterized by the dispatching of K models to clients and the uploading of K models from clients, leading to a communication overhead of $2K$.

Furthermore, the time complexity and spatial complexity on different training models are presented in Table 8. For CNN, the total time complexity is the sum of the time complexity of all convolutional layers. The computational amount of each layer is related to the length of the input sequence n, the size of the convolutional kernel k_2, the number of input channels c_{in}, and the number of output channels c_{out}. The spatial complexity includes the number of parameters of the convolution kernel and the output feature map generated by each layer. For an LSTM model, the time complexity is mainly determined by the sequence length n, hidden state dimension h, and input feature dimension d. The spatial complexity includes the number of parameters of the convolution kernel and the output feature map generated by each layer. The time complexity and spatial complexity of a CNN-LSTM model are the synthesis of the CNN part and the LSTM part. Time complexity of an LSTM-Transformer model is a combination of the quadratic growth of the Transformer and the linear growth of the LSTM. Spatial complexity is high due to the attention matrix. Besides, CNN, LSTM, CNN-LSTM, and LSTM Transformer each have their characteristics and applicable scenarios. CNN excels in image processing, LSTM is good at handling sequential data, CNN-LSTM combines the advantages of both, and LSTM Transformer has higher parallelization capabilities and global information capture when dealing with long-sequence data.

6. Conclusion

In this study, we introduce a robust personalized federated learning-based framework, pFedCross, to address the challenging problem of Non-IID data and privacy preservation in intrusion detection systems. pFedCross enhances personalized local models by proportionally aggregating them with a collaborative model and the global model. Further, a gradient approximation α-fairness algorithm is presented to produce the global model. pFedCross acknowledges the unique convergence traits of each client, avoiding the simplistic approach of averaging local models, which allows for swift convergence to local optima. The global model, while maintaining a balance between system efficiency and fairness, retains the model's generalization. Comprehensive experiments on four public datasets have demonstrated that pFedCross outperforms comparative algorithms significantly in various aspects, including classification precision, model loss, and other key performance indicators. For future research, we will investigate the potential integration of pFedCross with diverse deep learning algorithms and other FL frameworks.

CRediT authorship contribution statement

Shimin Sun: Writing – review & editing, Writing – original draft, Project administration, Methodology. **Le Zhou:** Software, Resources, Project administration, Methodology. **Ze Wang:** Methodology, Investigation. **Li Han:** Resources, Project administration, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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